Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Warehouse Traffic Control: Distributed Robot Coordination | *JavaScript, Three.js, Web Workers*

- Designed a lease-and-fencing protocol coordinating 200 robots across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Implemented Chandy–Misra–Haas deadlock detection and crash recovery that rebuilds a manager's state in ∼4.3 s.
- Built non-blocking gossip congestion control on lease replies, raising tasks completed by **98%** at 150 robots.
- Versus one central coordinator: **7.6x lower peak per-node load** (125 vs 950 msgs/s), and a region-manager crash cuts off **only 11 robots instead of all 120**.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling workers cut 1GB single-peer downloads from **193s to under 13s**.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

FOOL: Custom POSIX-Style Interactive Shell | *C++, POSIX APIs, Job Control, Signals, Linux*

- Built a C++ shell with N-stage pipelines, ordered descriptor redirection (`2>&1`) and full job control (process groups, terminal handoff, Ctrl+Z, `fg`/`bg`), verified by a **138-check** suite including real-PTY keystroke tests.
- Eliminated signal race conditions and lost exit statuses using the self-pipe trick for async-safe signal handling and centralizing process reaping on the main thread.
- Cut startup **52x** (5.2s to 0.1s) by reading file types from directory listings instead of checking every `PATH` entry individually; built a zero-dependency raw-mode line editor with tab autocompletion and persistent history.

RookDB: Vectorized Relational Storage Engine | *Rust, Database Internals, B+Tree, CBO, Volcano*

- Co-architected a multi-crate SQL-99 RDBMS in Rust verified by **700+ value-level tests**, engineering slotted-page storage, CLOCK buffer pools, B+Tree indexing, and Volcano query execution.
- Engineered vectorized Volcano query operators, a Cost-Based Optimizer, and Grace Hash Join, executing **2M-match joins in 5.85s** and streaming scans at **16.9M rows/s**.
- Designed an executor-tier cache and LRU query plan cache, eliminating 8 per-row I/O bottlenecks and driving a **100–400x DML throughput turnaround** (38,986 inserts/sec vs. 94 baseline) at 10M-row scales.

Technical Skills

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.

Systems & OS: POSIX APIs, Sockets, Linux/Unix, Multithreading, Signals, Termios, OpenSSL, ASan/UBSan.

Distributed Systems & Storage: Hadoop Streaming, HDFS/YARN, gRPC, Protobuf, SLURM, Storage Engines, Volcano Model.

AI, Web & Tools: PyTorch, Transformers, Scikit-learn, Next.js, React, Convex, PostgreSQL, Git, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.